The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then A, B, C and D may be identified as:

(A) Na, H_2, NaOH and Zn
(B) K, H_2, KOH and Zn
(C) K, H_2, NaOH and Zn
(D) Ca, H_2, $Ca(OH)_2$ and Zn

Sol^n:

$$\underset{(A)}{Na} + H_2O(l) \longrightarrow \underset{(C)}{NaOH(aq)} + \underset{(B)}{H_2\uparrow}$$

$$NaOH \xrightarrow{\underset{(D)}{Zn}} Na_2ZnO_2(aq) + \underset{(B)}{H_2\uparrow}$$

$$Zn \xrightarrow{\text{dil } H_2SO_4} H_2\uparrow + ZnSO_4(aq)$$

Calcium imide on hydrolysis gives gas (B) which on oxidation by bleaching powder gives gas (C). Gas(C) on reaction with magnesium give compound (D) which on hydrolysis gives again gas (B). Identify (B), (C) and (D).

(A) NH_3, N_2, Mg_3N_2

(B) N_2, NH_3, $MgNH$

(C) N_2, N_2O_5, $Mg(NO_3)_2$

(D) NH_3, NO_2, $Mg(NO_2)_2$

Sol^n:

Imp

NH_2^- amide

N^{3-}

NH^{2-} : imide.

H^-, B^{3-},

O_2^-, O_2^{2-}, O_3^-, CH_3^-

Stronger base than OH^- so fetch H^+ from water

$$CaNH \xrightarrow{+H_2O(l)} NH_3\uparrow + Ca(OH)_{2(aq)}$$

$$NH_3 \xrightarrow{CaOCl_2} \underset{(C)}{N_2} + CaCl_2$$

$$N_2 \xrightarrow{Mg} \underset{(D)}{Mg_3N_2}$$

$$Mg_3N_2 \xrightarrow{} \underset{(B)}{NH_3}\uparrow$$

$(Si_2O_5)_n^{2n-}$ anion is obtained when:

(A) no oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron

(B) one oxygen of a SiO_4^{4-} tetrahedron is shared with another SiO_4^{4-} tetrahedron

(C) two oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

(D) three oxygen of a SiO_4^{4-} tetrahedron are shared with another SiO_4^{4-} tetrahedron

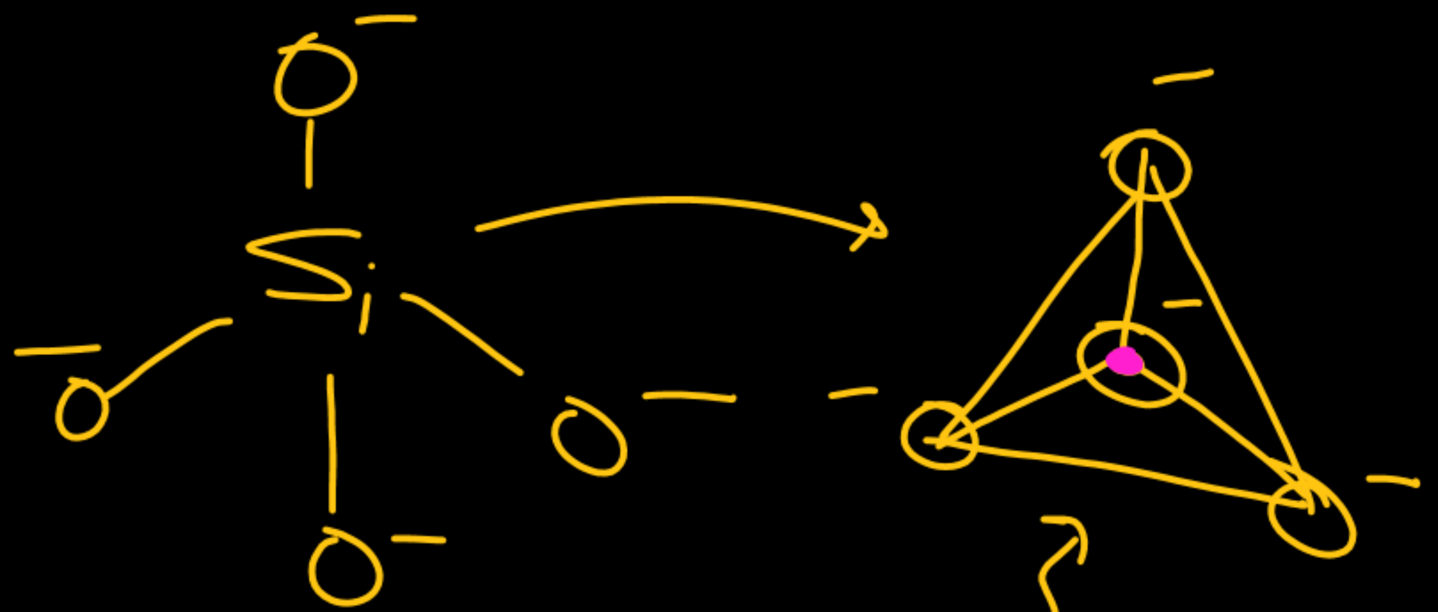
SiO_4^{4-} (ortho silicate)

Pyro silicate $Si_2O_7^{6-}$

chain/cyclic silicate $(SiO_3)_n^{2n-}$

Sheet silicates $(Si_2O_5)_n^{2n-}$.

Sol^n:



Tetrahedral silicate

2D representation

• : Silicon

O : oxygen

Which of the following reactions cannot result into adduct formation?

(A) $BF_3 + NH_3 \rightarrow$

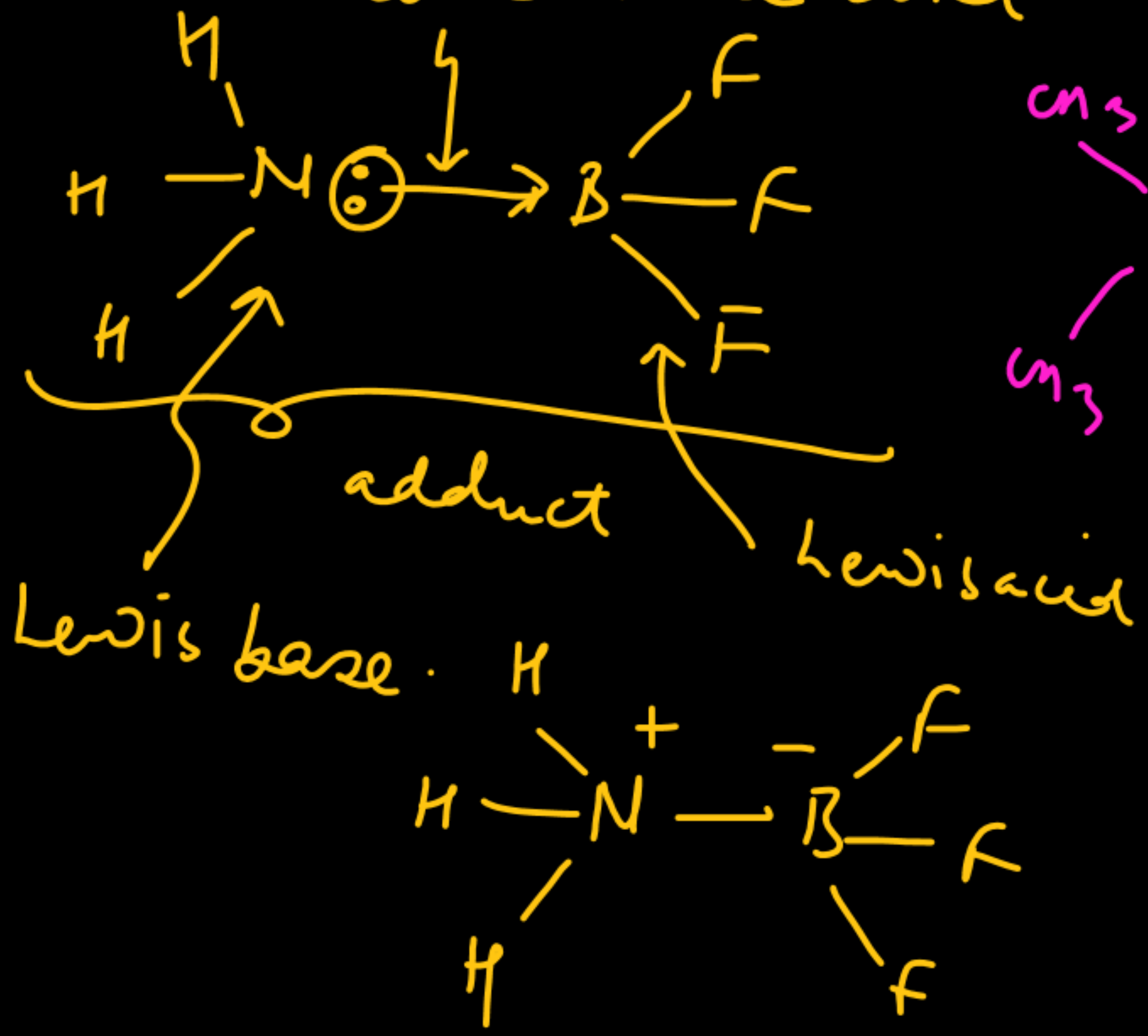
(B) $BCl_3 + NH_3 \rightarrow$

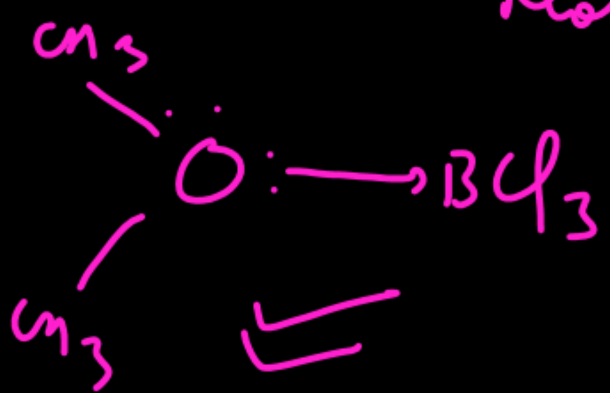
(C) $CH_3 - O - \overset{CH_3}{NH_3} + BCl_3 \rightarrow$

(D) $H_3Si - O - SiH_3 + BCl_3 \rightarrow$

Not a Lewis base.

addition product.





Disilyl ether,

$\theta \cong (130 \text{ to } 160^\circ)$

$p\pi - d\pi$ back bond.

does not act as good L.B.

Match the reactions listed in column-I with characteristic(s) / type of reactions listed in column-II.

Column-I		Column-II
(A) $BBr_3 + H_2 \rightarrow B$	q	(p) Borax bead test
(B) $Na_2B_4O_7.10H_2O + CuSO_4 \rightarrow Cu(BO_2)_2$	p	(q) Reduction
(C) $AlCl_3 + H_2O \rightarrow HCl + Al(OH)_3\downarrow$	r, s	(r) White fumes
(D) $Cr_2O_3 + Al \rightarrow Cr + Al_2O_3$	q	(s) Hydrolysis

(A) (A) – q; (B) – p; (C) – r, s; (D) – q ✓

(B) (A) – p; (B) – q; (C) – r, s; (D) – s

(C) (A) – q; (B) – p; (C) – p, s; (D) – r

(D) (A) – p; (B) – r; (C) – q, r; (D) – s

Solⁿ: HCl has strong affinity to absorb moisture and becomes cloudy.

Incorrect statement is :

(A) $NaHCO_3$ and $KHCO_3$ have same crystal structure

(B) On heating $LiNO_3$ decomposes into Li_2O, O_2 and NO_2

(C) Among alkali metals, Li metal impart red colour to flame

(D) Li_2SO_4 does not form alum

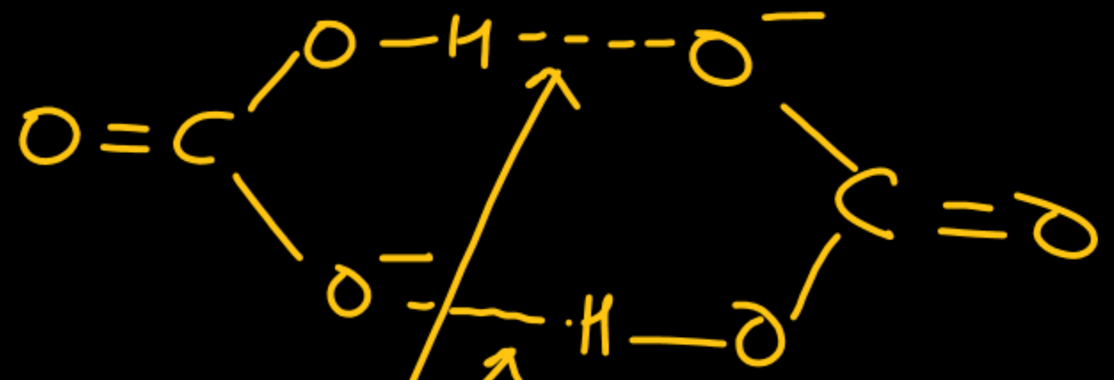
Li Be B

Na Mg Al C

Imp

$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2\uparrow + O_2\uparrow$

Solⁿ : $KHCO_3$ forms H-bonded dimer.



H-bonding

dimeric

$NaHCO_3$

HCO_3^- are polymerised through H-bonding.

A complex cross-linked polymer (silicone) is formed by

(A) hydrolysis of $(CH_3)_3$ SiCl.

(B) hydrolysis of a mixture of $(CH_3)_3SiCl$ and $(CH_3)_2SiCl_2$

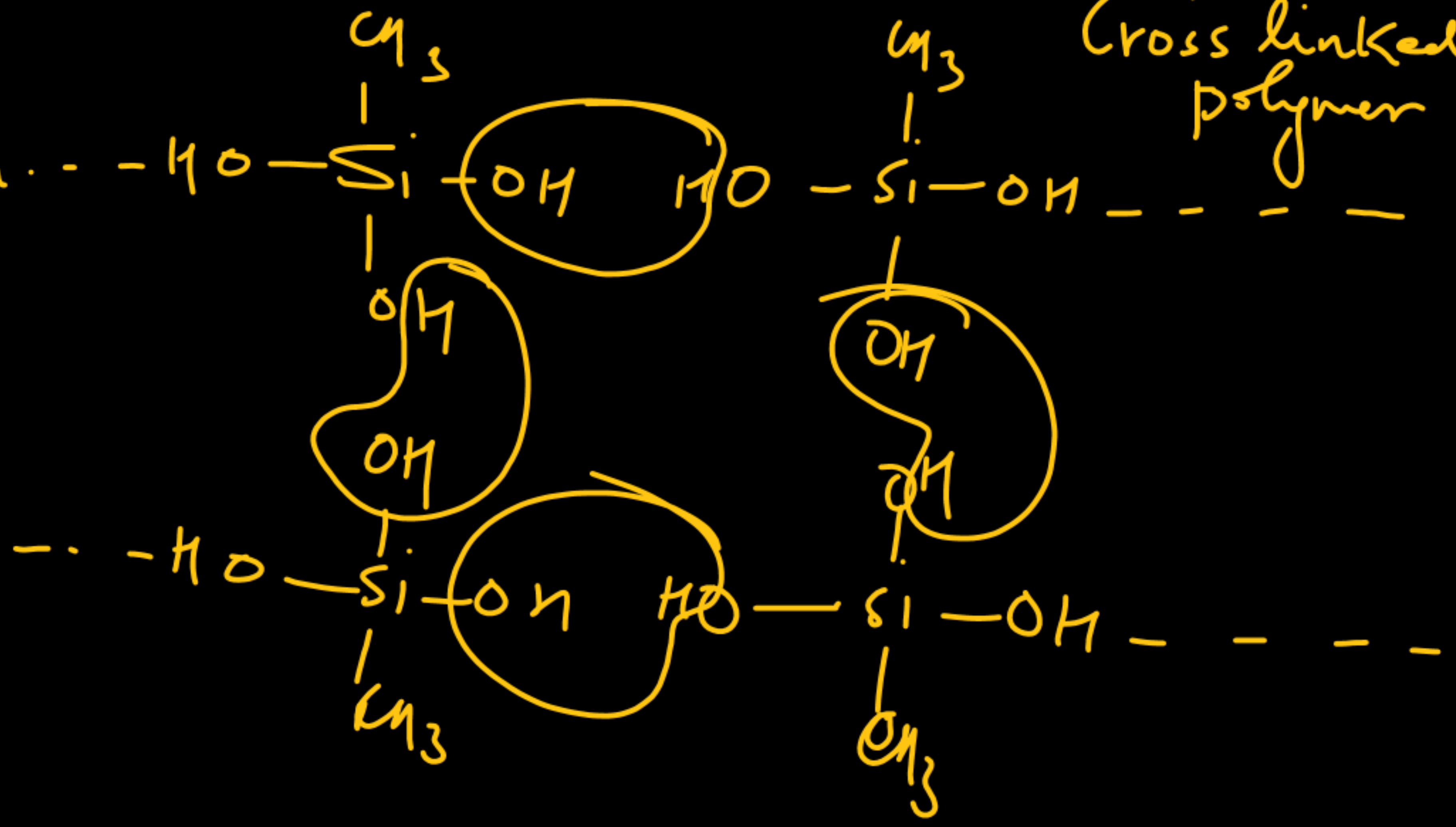
(C) hydrolysis of CH_3 $SiCl_3$

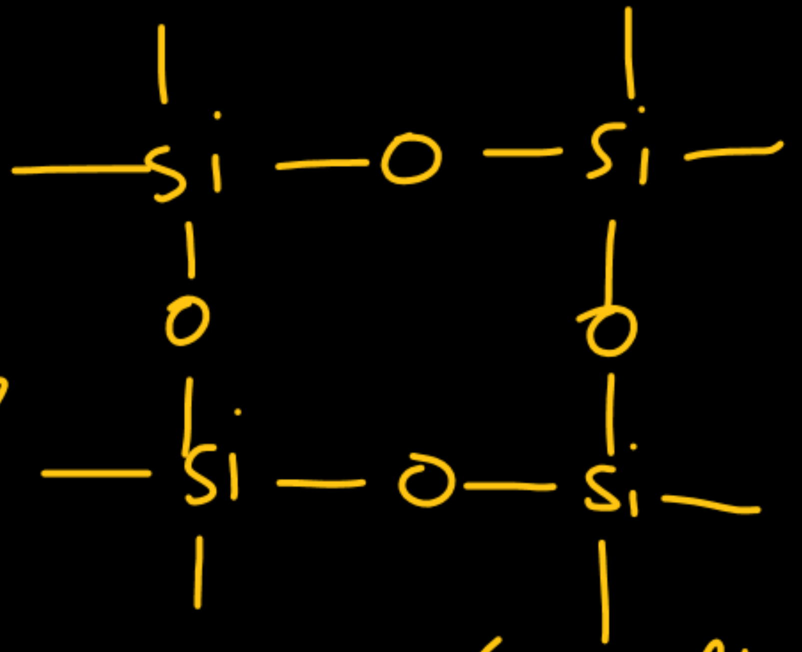
(D) hydrolysis of $SiCl_4$.

Imp

NCERT

$CH_3SiCl_3 \xrightarrow[\text{(Condensation)}]{\text{hydrolysis}}$





Cross linked polymer.

Imp

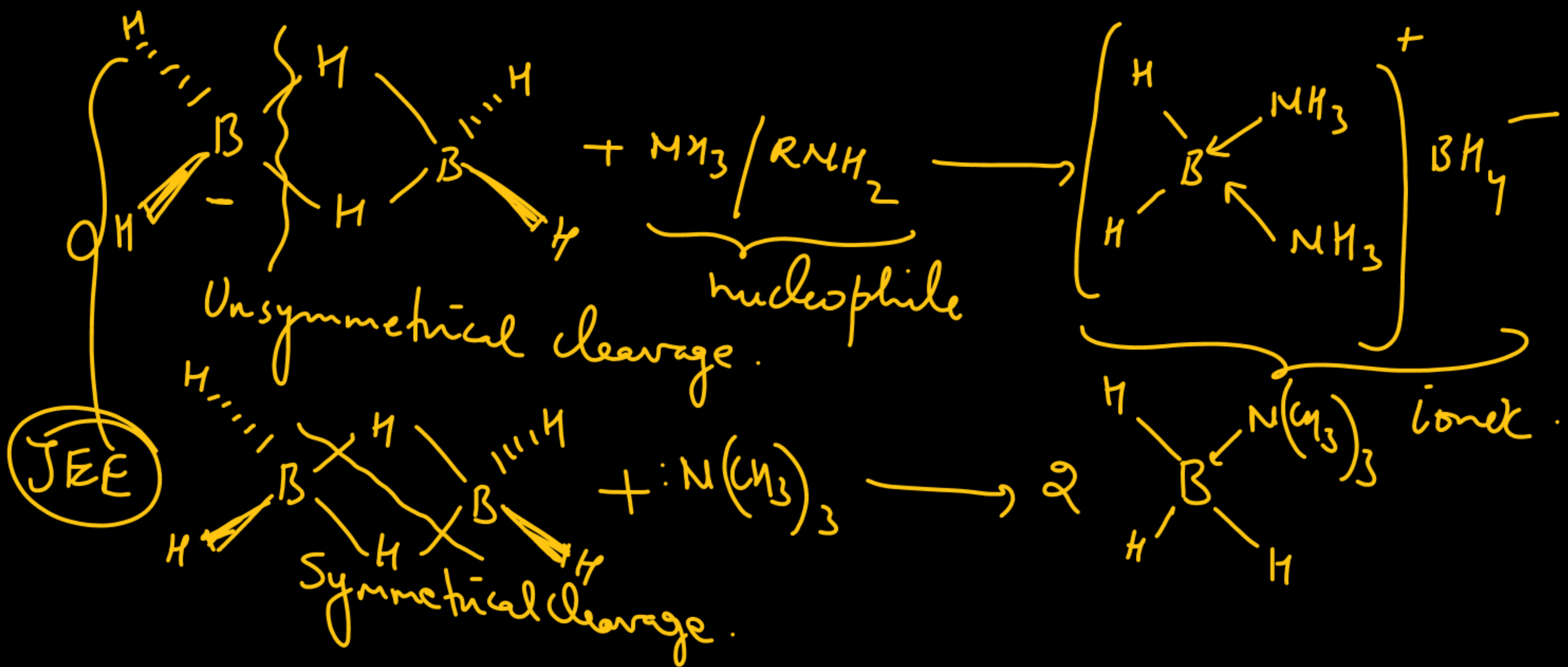
Which of the following statement is correct for diborane ?

(A) Small amines like NH_3, CH_3NH_2 give unsymmetrical cleavage of diborane.

(B) Large amines such as $(CH_3)_3N$ and pyridine gives symmetrical cleavage of diborane.

(C) Small as well as large amines both gives symmetrical cleavage of diborane.

(D) (A) and (B) both



Imp

Which reaction cannot give anhydrous $AlCl_3$?

(A) Heating of $AlCl_3.6H_2O$

(B) Passing dry HCl over heated aluminium powder

(C) Passing dry Cl_2 over heated aluminium powder

(D) Heating a mixture of alumina and coke in a current of dry Cl_2

Solⁿ: $AlCl_3 \cdot 6H_2O \xrightarrow{\Delta} Al_2O_3(s) + H_2O\uparrow + HCl\uparrow$

hydrated aluminium chloride

hydrolysed product

(Internal hydrolysis)

$Al_{(s)} + \text{dry } HCl \rightarrow AlCl_3 + H_2\uparrow$

heated

$Al_{(s)} + Cl_2 \rightarrow AlCl_3$

(OA)

Stubborn

$Al_2O_3 / BeO / B_2O_3 + C + Cl_2 \xrightarrow{\Delta} AlCl_3 / BeCl_2 / BCl_3 + CO\uparrow$

Imp

Select correct statement(s)

(A) Borax is used as a buffer

(B) 1 M borax solution reacts with equal volumes of 2 M HCl solution

(C) In borax, boron is both sp^2 & sp^3 hybridised

(D) Coloured bead obtained in borax-bead test contains metaborate.

Ans : A, B, C, D

$2M \times V = 2V$

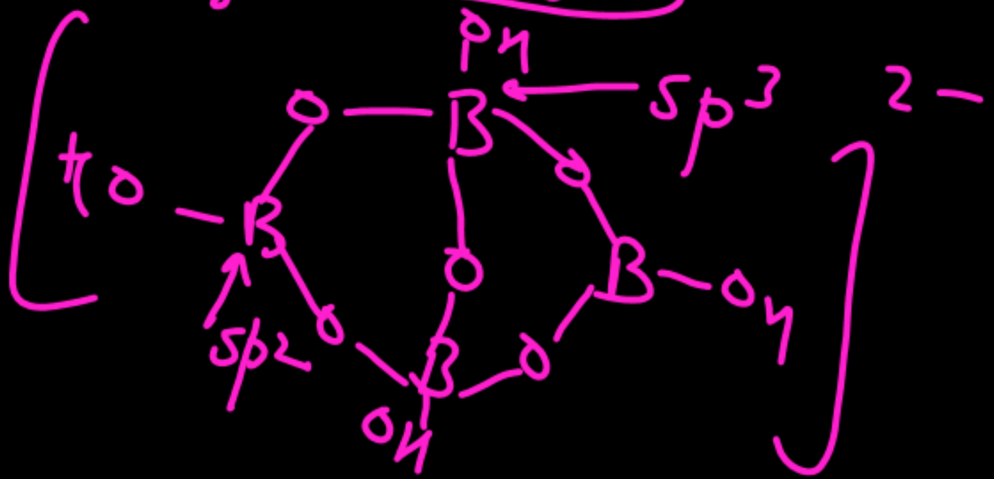
Sol^n: $Na_2B_4O_7 \cdot 10H_2O \xrightarrow[\text{(excess)}]{+H_2O(l)} \underbrace{2B(OH)_3}_{\text{Boric acid}} + 2[B(OH)_4]^-$ conjugate base

1M, V lit = 1V moles

$2B(OH)_3$: 2V mol $\xrightarrow{+HCl}$ No rex

$[B(OH)_4]^-$: 2V moles $\xrightarrow{+HCl \text{ (2V moles)}}$ $2B(OH)_3 + 2H_2O$

$2Na^+[B_4O_5(OH)_4]^{2-} \cdot 8H_2O$



Which statements are false ?

Ans : C, D

(A) Manufacture of NaOH is done by Solvay process

(B) Manufacture of K_2CO_3 is done by Solvay process

(C) Manufacture of NaOH is done by Castner Kelner process

(D) Manufacture of $NaHCO_3$ is done by Solvay process

$NH_3 + CO_2 + H_2O(l) \longrightarrow NH_4CO_3$

N. Imp

$\downarrow + KCl_{(aq)}$.

$KHCO_3(aq)$
soluble
(No ppt).

for $NaHCO_3$

→ Solvay's process (also called Ammonia soda process)

$NH_3 + CO_2 + H_2O \rightarrow NH_4HCO_3$.

$$\underset{\text{Brine}}{NaCl_{(aq)}} + NH_4HCO_3 \xrightarrow[t<30^\circ C]{\text{Cool}} \underset{\text{ppt}}{NaHCO_3\downarrow} + NH_4Cl_{(aq)}.$$

$$\xrightarrow{\Delta} Na_2CO_3(s) + H_2O\uparrow + CO_2\uparrow.$$

Select the wrong statement (s) :

Ans : A, B

(A) CaF_2 is soluble in water

(B) $BaSO_4$ is soluble in water

(C) $Ba(OH)_2$ is soluble in water

(D) $MgSO_4$ is soluble in water

Sol^n :

Baryta water, $Ba(OH)_{2(aq)}$

(water soluble)

(i) A white precipitate (B) is formed when a mineral (A) is boiled with Na_2CO_3 solution.

(ii) The precipitate is filtered and filtrate contains two compounds (C) and (D). The compound (C) is removed by crystallisation and when CO_2 is passed through the mother liquor left (D) changes to (C).

(iii) The compound (C) on strong heating gives two compounds (D) and (E).

(iv) (E) on heating with cobalt oxide produces blue coloured substances (F).

Identify (A) to (F) and gives chemical equations for the reactions at steps (i) to (iv).

Sol^n:

Imp

$Ca_2B_6O_{11}\cdot 5H_2O$ → Colemanite ore.

$$\underset{(A)}{Ca_2B_6O_{11}\cdot 5H_2O} + Na_2CO_3(aq) \xrightarrow[\Delta]{Boil} \underset{(B)}{CaCO_3\downarrow} + \underset{(C)}{Na_2B_4O_7(aq)} + \underset{(D)}{NaBO_2(aq)}$$

$$Na_2B_4O_7(aq) \rightarrow Na_2B_4O_7\cdot 10H_2O \text{ (crystallised)}.$$

$$NaBO_2(aq) + CO_2(g) \rightarrow \underset{(C)}{Na_2B_4O_7\cdot 10H_2O} + Na_2CO_3$$

$$Na_2B_4O_7 \xrightarrow{\Delta} \underset{(D)}{NaBO_2} + \underset{(E)}{B_2O_3}$$

$$B_2O_3 \xrightarrow{CoO} Co(BO_2)_2 \text{ Blue Bead.}$$

A → $Ca_2B_6O_{11}\cdot 5H_2O$

B → $CaCO_3$

C → $Na_2B_4O_7$

D → $NaBO_2$

E → B_2O_3

F → $Co(BO_2)_2$

Carbon monoxide is prepared by :

(A) heating formic acid with conc. H_2SO_4

(B) heating potassium ferrocyanide with conc H_2SO_4

(C) heating malonic acid with P_4O_{10}

(D) hydrolysis of Mg_2C_3

Solⁿ :

$HCOOH \xrightarrow[H_2SO_4]{Conc.} CO\uparrow$

$K_4[Fe(CN)_6] \xrightarrow[H_2SO_4]{Conc.} CO\uparrow$

$CH_2(COOH)_2 \xrightarrow[\text{(Powerful dehydrant)}]{P_4O_{10}} C_3O_2$ (Carbon suboxide)

$O=C=C=C=O$ — sp

$Mg_2C_3 \xrightarrow{+H_2O(l)} CH_3-C\equiv CH$ (1-propyne)

$2Mg^{2+}$ C_3^{4-} allylide ion

Match the reactions listed in column-I with characteristic(s) / type of reactions listed in column-II.

Column - I		Column- II
(A) $Al_2(C_2)_3 + H_2O \longrightarrow$	C_2H_2 p, q	(p) One of the products contains both σ and π bonds
(B) $CH_2(COOH)_2 + P_4O_{10} \longrightarrow$	C_3O_2 p, r	(q) Hydrolysis
(C) $CH_3SiCl_3 + H_2O \longrightarrow$	s, q	(r) Dehydration
(D) $SnCl_2.2H_2O \xrightarrow[\text{standing}]{\text{on}}$	$Sn(OH)Cl$ q	(s) complex crosslinked polymer

C^{4-}

C_2^{2-}

C_3^{4-}

Sol^n: (A) $Al_2(C_2)_3 \xrightarrow{+H_2O(l)} Al(OH)_3\downarrow + C_2H_2 \Rightarrow H-C\equiv C-H$

1σ
2π

$2Al^{3+}$ $3C_2^{2-}$

(acetylide)

Write balanced equations for the following reactions :

(A) SnO is treated with dilute HNO_3

(B) Tin is treated with an excess of chlorine gas.

(C) Lead sulphide is heated in air.

Compound (X) on reduction with $LiAlH_4$ gives a hydride (Y) containing 21.72% hydrogen along with other products. The compound (Y) reacts with air explosively resulting in boron trioxide. Identify (X) and (Y). Give balanced reactions involved in the formation of (Y) and its reaction with air. Draw structure of (Y).